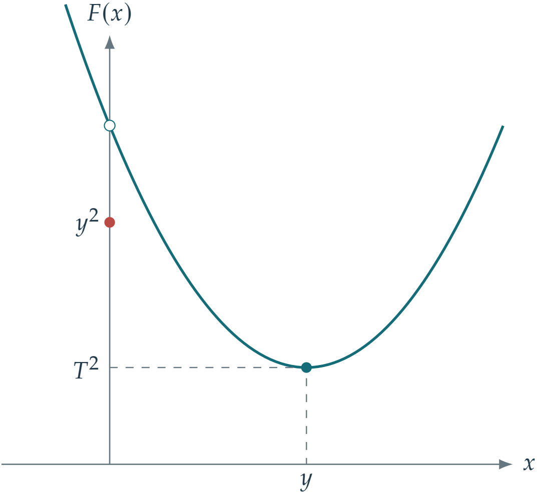

$$F(x) = (x-y)^2 + T^2 \mathbf{1}_{x \neq 0}$$

$$|y| < T: \quad x^* = 0$$

$$|y| > T: \quad x^* = y$$

$$|y| = T: \quad x^* \in \{0, y\}$$

Shown: $y > T > 0$.